Grey goo as a technosignature: propagation timescales from planet to cosmos, the observational record, and what its silence does and does not say about our own risk

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Abstract

"Grey goo"—a self-replicating machine population that has escaped every control on its growth—has been discussed for forty years as a planetary catastrophe and, separately, self-replicating probes have been discussed for fifty as a way of exploring the Galaxy. The two literatures have not met: no published work asks how far an *uncontrolled* replicator propagates, on what timescale, what it would look like from outside, and what the fact that we do not see it implies for the probability that a technological civilisation destroys itself this way. We supply the missing calculation. We separate the question by scale and by the one property that governs it, dispersal. In place on a planet, replication is never the bottleneck: a biosphere is consumed in hours to days at any plausible doubling time, but a crust takes $\sim$4.5 kyr and a bulk planet $\sim$1.1 Myr because the surface cannot shed the heat of processing faster than $4\pi R^2 \sigma T^4$ at the temperature the machines survive. In a planetary system the small bodies are transport-limited (14 yr for the asteroid belt) and produce a Dyson-scale waste-heat source at the feedstock's orbital temperature (169 K for the belt); we show that a *detectable* dead residue is necessarily a *short-lived* one, surviving at most $P_{\mathrm{orb}}/f_{\min}$ above the detection floor $f_{\min}$ (4.4 kyr at 2.7 AU for $f_{\min} = 10^{-3}$), so that the existing waste-heat surveys constrain live goo and barely constrain dead goo. Between stars there are two roads. A replicator never designed to travel still leaves its system by radiation-pressure blowout if it is smaller than 0.57 μm, and the same radiation pressure lets only the 0.21–0.57 μm size class reach the habitable zone of the next star; a single asteroid belt's worth of such machines, phase-mixed around the solar annulus in $\sim$1.2 Gyr, delivers of order 200 intact devices per year to every rocky planet in the annulus before survival losses, so that if a machine can stay functional in the interstellar medium for $\gtrsim 10^8$ yr the passive branch seeds a galactic annulus in a gigayear without any spacecraft. A contact-graph model of every natural channel that moves mass between systems (interstellar objects, impact ejecta, free-floating planets, stellar flybys, interstellar dust, supernova ejecta, birth-cluster exchange), with every rate from the literature, shows that the Galaxy is already completely connected in the mass sense—every Earth-sized planet has been struck tens of times by bodies formed around other stars, and one system's ejected planetesimals are captured by 7.6×10^6 other systems—but sits at the percolation threshold ($R_0 \approx 2.8$) for one system's material landing on other planets, so that contact is never the bottleneck for a replicator loose among a system's small bodies and never the route for one confined to a planet. The active branch—a replicator that inherits interstellar propulsion—fills the Galaxy in 639 kyr–504 Myr and reaches 7×10^8–5×10^9 galaxies before the cosmic event horizon closes at 16.6 Gly. We compile what existing searches say at each scale and show that the constraints separate cleanly into three tiers: our own system's intact belt and living biosphere (anthropically shadowed), the $\sim 5 \times 10^6$-star waste-heat surveys (unshadowed but short-memoried), and the 10^5-galaxy $\hat{\mathrm{G}}$ survey (unshadowed and long-memoried). A Bayesian treatment with and without the anthropic shadow gives, per civilisation, $p \lesssim 3.0 \times 10^{-7}$ for a persistent galaxy-spanning goo if $\Lambda \sim 100$

civilisations have arisen per galaxy, but *no* constraint on the planet-bound branch at any Λ. Mapped back onto an elicited planet-scale accident probability, the astronomical record moves it by a factor 0.990 if one per cent of such accidents escape their system, and by at most 0.50 if half do. The silence is informative about goo that travels and silent about goo that stays home; the goo we can accidentally make is, for the next several centuries, the kind that stays home. We state what observation would change this and generalise the argument to the other ways a civilisation can die of its own replicators.

1 Introduction

The phrase "grey goo" was coined by Drexler [1986] for a thought experiment: a molecular assembler that copies itself from ambient matter, escapes whatever was meant to stop it, and converts the biosphere into more of itself. The scenario acquired quantitative form in Freitas [2000], who bounded the rate of "global ecophagy" by the heat the replicators must shed, and was then disowned by its author on engineering grounds [Phoenix and Drexler, 2004]: self-replication is not needed for manufacturing, and a replicator able to eat an unprepared environment is a much harder thing to build than one that runs in a vat. The scenario nonetheless remains a standing entry in the existential-risk literature [Bostrom, 2002, Rees, 2003, Ord, 2020], and every survey of that literature assigns it a small but non-zero probability.

A separate literature, older and larger, treats self-replicating machines in space as a good idea. Tipler [1980] argued from von Neumann probes that a single civilisation could occupy the Galaxy in $\lesssim 3 \times 10^8$ yr and that the absence of such probes here shows there is no one else; Freitas [1980] designed one; Hart [1975], Newman and Sagan [1981], Landis [1998], Bjørk [2007], Cotta and Morales [2009], Wiley [2011], Nicholson and Forgan [2013] and Carroll-Nellenback et al. [2019] computed the colonisation front in increasing detail; Armstrong and Sandberg [2013] and Olson [2015] carried it to intergalactic scales, and Hanson et al. [2021] built a cosmological model of "grabby" expansion on it. Within that literature the one paper that reads like a goo argument is Sagan and Newman [1983], who held that a self-reproducing probe population would become a danger to its own makers and would therefore be suppressed by any prudent civilisation, a position Tipler rejected.

What neither literature contains is the calculation the title of this paper names. The planetary goo literature stops at the planet; the probe literature assumes design, purpose and control. The nearest approaches are Stevens et al. [2016], who list grey goo among the ways a civilisation might destroy itself and sketch its planetary signature (a dead, chemically altered surface, perhaps with an artificial spectral feature), and the recent papers on self-replicating probes as solar-system technosignatures [Ellery, 2022a,b, 2025], which are concerned with designed probes and their curbing. Nobody has asked how far an *uncontrolled* replicator can go, how long it takes to get there, what it would look like at each scale, whether the data already exclude it, and—the question with a practical stake—whether the answer should move our estimate of the probability that *we* do it.

We answer these questions in order. Section 2 fixes what we mean by goo and identifies dispersal, not replication rate, as the property that sets its reach. Sections 3–6 compute the timescales at the planetary, system, galactic-passive, galactic-active and cosmological scales, and in each case say what the residue looks like and how long it lasts. Section 7 widens the passive branch into a contact-graph model of every natural channel by which mass moves between systems, and asks how long the Galaxy takes to be touched by something touched by something else. Section 8 compiles the observational record. Section 9 does the inference, with and without the anthropic shadow, and maps it onto our own risk. Section 10 generalises to the other self-replicating technologies with which a civilisation could kill itself, and Section 11 states the assumptions on which the argument turns and the observations that would change it. Every number in the paper is generated by the released code from a single parameter file, and where a parameter is uncertain by orders of magnitude we scan it rather than choose it.

2 What grey goo is, physically

We take a replicator to be any system that builds functional copies of itself from its surroundings. It is characterised by four numbers: the doubling time τ under favourable conditions; the energy e_{rep} consumed per unit mass of replicator built; the range of feedstock it can use, which we do not model but which enters every result as an assumption of *generality*; and the mass m of the unit, which through its size s sets how it moves under gravity and light. We use $\tau = 10^3$ s, $e_{\mathrm{rep}} = 1 \times 10^7\,\mathrm{J\,kg^{-1}}$ (one electron-volt per atom placed), $s = 0.5\,\mu$m and $\rho = 2\,\mathrm{g\,cm^{-3}}$ as fiducial, and scan τ from 100 s to a year. The fiducial device has mass 1.0×10^{-15} kg; from one such seed a planet's mass is 132 doublings away.

"Goo" is a replicator population without an external control on its growth. It is not defined by malevolence, purpose or intelligence; a goo can be a manufacturing organism, a terraforming seed, an industrial swarm or a probe whose replication limiter failed. What it is defined by is that it keeps going until something physical stops it. Four things can: exhaustion of feedstock it can use, the heat of its own metabolism, the error catastrophe that afflicts any finite-fidelity copier [Eigen, 1971, Kowald, 2015], and predation or competition by other replicators [Chen et al., 2022]. We treat the first two quantitatively; the last two enter as a residue-lifetime parameter and as a caveat.

The property that determines how far a goo goes is none of these. It is *dispersal*: whether the units can leave the body they are on, the system they are in, and the galaxy. A goo that cannot leave its planet is a planetary catastrophe with no astronomical consequence; a goo that has space transport is a system-scale event; a goo whose units are small enough to be blown out of the system by starlight is an interstellar event on a gigayear timescale; a goo that inherited interstellar propulsion is a galactic event on a megayear timescale. Replication rate sets how fast each scale is eaten once reached, and we will find that it is almost never the bottleneck; dispersal sets which scales are reached at all. We therefore organise the paper by dispersal class, and the inference in Section 9 will turn on the fraction of goo accidents that fall into each.

3 Planet scale: the three ceilings

The number usually quoted for goo is the exponential one: $\log_2(M/m)$ doublings at τ each. For Earth's biosphere (1.1×10^{15} kg dry; Bar-On et al. 2018) that is 100 doublings and 28 h at $\tau = 10^3$ s; 2.8 h at 100 s, 101 yr at a year. For the crust it is 124 doublings and for the bulk planet 132; the extra 30 doublings cost a day. This is the source of the folk belief that goo eats worlds in days, and it is wrong beyond the biosphere for a reason Freitas [2000] identified for the biosphere itself: heat.

Every kilogram of replicator built dissipates e_{rep} as heat, whatever the energy source—sunlight, the free energy of the feedstock, or fission. A planetary surface at temperature T can shed at most $4\pi R^2\sigma(T^4 - T_{\mathrm{eq}}^4)$, which for Earth is 1.7×10^{18} W at 500 K and 4.6×10^{20} W at 2000 K. A machine that must stay below 500 K to function therefore cannot process more than $1.7 \times 10^{11}\,\mathrm{kg\,s^{-1}}$ planet-wide, and the crust takes 4.5 kyr, the bulk 1.1 Myr, at any τ. This radiative ceiling is independent of where the energy comes from; the second ceiling is where it comes from. Absorbed sunlight is 1.2×10^{17} W, which alone would stretch the crust to 62 kyr and the bulk to 16 Myr; the crust's uranium and thorium hold 8×10^{30} J and the oceans' deuterium 7×10^{31} J, together enough for the crust and marginally for the bulk, so that a goo with nuclear metabolism is radiatively rather

Table 1: In-place conversion of Earth's reservoirs at $\tau = 10^3\,\mathrm{s}$, $e_{\mathrm{rep}} = 1 \times 10^7\,\mathrm{J\,kg^{-1}}$, machines limited to 500 K. The governing time is the slowest ceiling. Raising the operating temperature to 2000 K shortens the crust to 16 yr and the bulk to 4.1 kyr, which is why a goo that can run hot is a qualitatively worse thing.

reservoir	doublings	exponential	radiative (500 K)	solar-powered
biosphere (1.1×10^{15} kg)	100	28 h	1.8 h	25 h
crust (2.4×10^{22} kg)	124	~1.5 d	4.5 kyr	62 kyr
bulk (6.0×10^{24} kg)	132	~1.5 d	1.1 Myr	16 Myr

than solar-limited. The governing timescale is the slowest of the three (Table 1): hours for the biosphere, 4.5 kyr for the crust, 1.1 Myr for the planet, the last two independent of τ from 100 s to a year.

Two remarks. First, these are conversions in place; *lifting* the mass off the planet is a different budget, set by the gravitational binding energy $3GM^2/5R$: a week at the full solar luminosity (7 d, Dyson's famous number), 15 kyr at the 2000 K surface ceiling, 58 Myr on absorbed sunlight alone. A goo confined to a planet does not dismantle it into a swarm; it converts it into a planet made of goo. Second, Freitas [2000] reached a similar conclusion for the biosphere by a different route—he bounded the global replicator power at $\sim 10^{13}$ W by the warming it would cause and found ecophagy times of order a year or more at that limit, and argued that the thermal signature of a replicating patch would be detectable from orbit well before it was global. Our ceilings are the physical ones a goo indifferent to the climate would hit; his is the one a defender could use.

What it looks like from outside. A planet during ecophagy re-radiates the energy it always did plus the metabolism of the goo, which at the radiative ceiling is a ~ 500 K surface—a hot rocky planet with the wrong equilibrium temperature for its orbit. Afterwards it is a dead world with a refined surface: metallic or graphitic, uniform, without the disequilibrium chemistry of a biosphere [Stevens et al., 2016]. Nothing in the current exoplanet census can see either. Thermal emission from rocky planets has been measured for a handful of hot, close-in worlds [Kreidberg et al., 2019, Greene et al., 2023, Zieba et al., 2023] and reflects bare rock; surface composition of a temperate planet is beyond any funded facility. The planetary branch is therefore unobservable by construction for the next decades, and this will matter in Section 9.

4 System scale: the belt, the planets, and the residue

A goo with any space transport—it ate a spacefaring civilisation's infrastructure, or was one—faces a different problem: the feedstock is in 10^6 bodies spread through a torus. Inside a belt the front is a branching process. Each converted body launches b packets to its neighbours; a hop covers the mean spacing at a Δv of a kilometre per second and waits an orbital period for phasing; a body saturates after ~ 100 local doublings. With $b = 100$ the 10^6-body main belt is consumed in three generations, 14 yr, and the 10^9-body Kuiper belt in 1.1 kyr—limited by transport and orbital phasing, not by τ, which it hardly sees. Planets again cost more: their mass leaves at the rate the surface can shed the heat, 7.7 kyr for a terrestrial planet and 143 kyr for a giant at the 2000 K ceiling (52 h and 21 yr if the full stellar luminosity could be delivered to the job, which it cannot). The system is therefore eaten from the outside in, small bodies first, and Figure 1 shows the whole ladder.

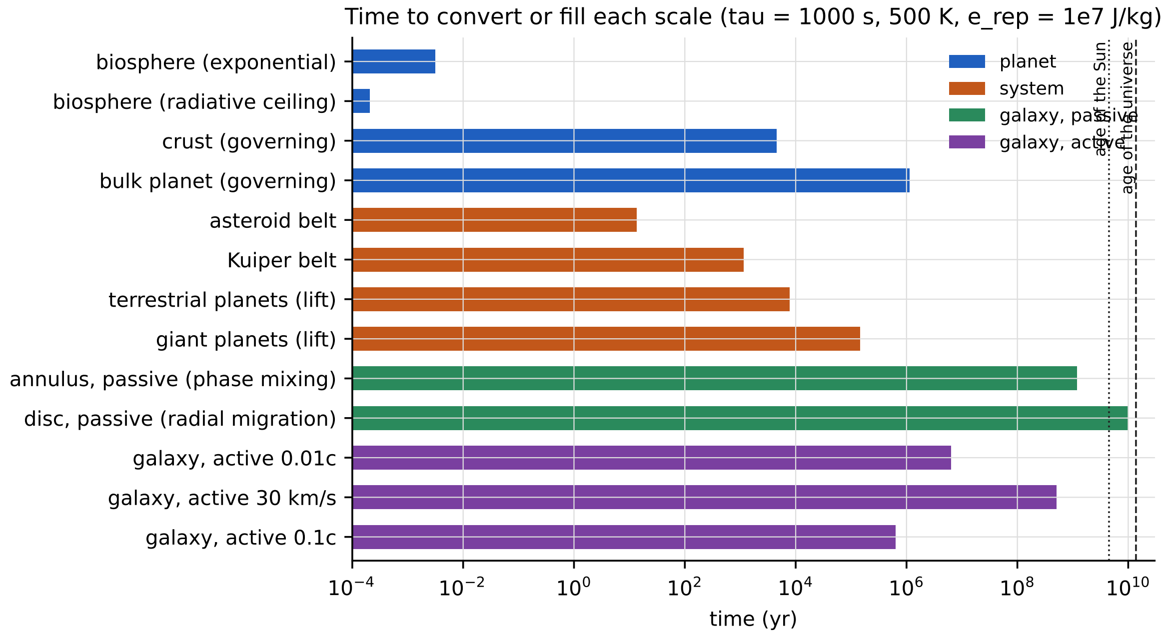

Figure 1: Time to convert or fill each scale, by dispersal class. Planetary conversion is set by the radiative ceiling; system-scale conversion by transport; the passive interstellar branch by galactic phase mixing; the active branch by ship speed.

The residue. The converted mass is a swarm. Spread at areal density Σ on a shell at radius r it covers a fraction $f = \min(1, M/4\pi r^2\Sigma)$ of the star; the main belt's 2.4×10^{21} kg covers the star completely as micron-thick devices ($\Sigma = 10^{-3}\,\mathrm{kg\,m^{-2}}$), a fraction 0.0012 as kilogram-per-square-metre sheets, and 1.2×10^{-6} as metre-scale slabs. A thin absorber at r radiates at $T = 278\,\mathrm{K}\,(r/\mathrm{AU})^{-1/2}$: 169 K for the belt, 44 K for the Kuiper belt. A live system-scale goo is therefore a partial Dyson sphere at the temperature of its feedstock's orbit, and every waste-heat search ever run [Carrigan, 2009, Wright et al., 2014b,a, Suazo et al., 2022, 2024] is a search for it, with one difference in expectation: a purposeful swarm sits where its builders want the energy, a goo sits where the rock was.

A dead one does not last. Lacki [2025] showed that an unmaintained swarm of covering fraction f grinds itself down on a collisional time that "is roughly an orbital period divided by the covering fraction of the swarm", $\sim P_{\mathrm{orb}}/f$, so that a fully randomised swarm "self-destructs within a year or so"; the fragments then leave by Poynting–Robertson drag [Burns et al., 1979] on $400\,\mathrm{yr}\,(r/\mathrm{AU})^2/\beta$ or, below the blowout size, on one orbit. For the belt mass as 0.5 μm machines the residue is gone in 4.4 yr; as millimetre grains ($f = 4.4 \times 10^{-4}$) in 10 kyr; as metre slabs ($f = 4.4 \times 10^{-7}$) it lasts 10 Myr but is far below any detection floor. This is a general statement, and it is the one that governs what the surveys can say: a residue detectable as an infrared excess needs $f \geq f_{\mathrm{min}}$, and its collisional lifetime is then at most

$$t_{\mathrm{vis}} \leq \frac{P_{\mathrm{orb}}}{f_{\mathrm{min}}}, \tag{1}$$

which at 2.7 AU is 4.4 kyr for $f_{\mathrm{min}} = 10^{-3}$, 444 yr for 10^{-2} and 44 yr for 10^{-1} (25 kyr at 40 AU for 10^{-2}). *A detectable dead residue is a short-lived one.* Figure 2 draws the consequence. The only long-lived system-scale signature is a goo that is still alive—still replicating on its own collision fragments, still re-radiating its captured starlight—and that is what the Dyson searches bound.

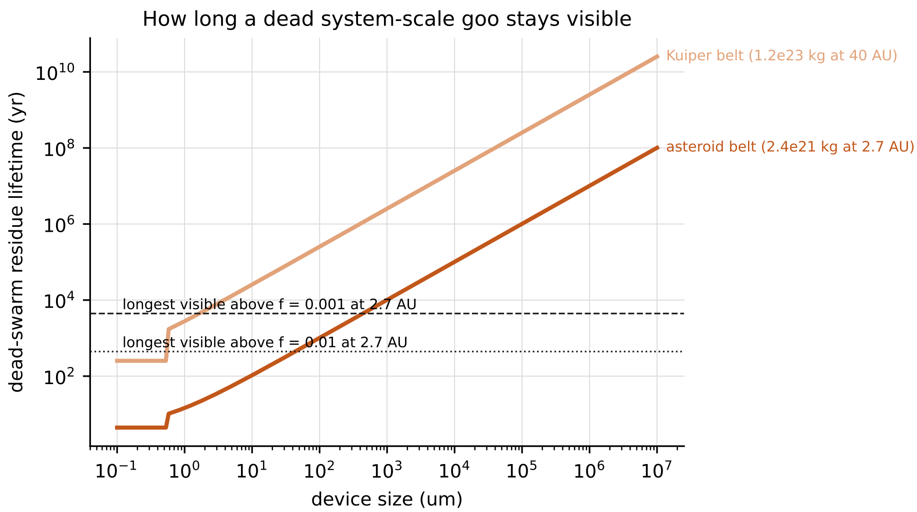

Figure 2: Lifetime of a dead swarm as a function of the size of the bodies it is ground into, for the main-belt mass at 2.7 AU and the Kuiper-belt mass at 40 AU, against the longest a residue can stay above an infrared-excess detection floor (equation 1). Sub-blowout devices leave on an orbit; larger ones on the shorter of the collisional and Poynting–Robertson times.

5 Between stars, passively: goo as interstellar dust

A replicator never designed to travel has one road out of its system, and it is the road interstellar dust takes. Around the Sun the ratio of radiation pressure to gravity on a sphere is $\beta = 0.57\, Q_{\rm pr}/(\rho\, s)$ with ρ in $\mathrm{g\,cm^{-3}}$ and s in microns [Burns et al., 1979]; devices smaller than 0.57 μm at $\rho = 2$ have $\beta > 0.5$ and are unbound the moment they are released from an orbiting body, leaving on a hyperbola with $v_\infty = v_{\rm circ}\sqrt{2\beta - 1}$: 7 $\mathrm{km\,s^{-1}}$ for our fiducial device from 2.7 AU, 18 $\mathrm{km\,s^{-1}}$ at $\beta = 1$. A goo that grinds an asteroid belt into micron-scale machines therefore ejects the sub-blowout part of its own mass at stellar speeds, at no cost and without intent. (Devices bound in a swarm also leave whenever a collision frees them; devices on a planet's surface do not, except as impact ejecta, a channel we return to below.)

The same force governs arrival. An incoming device of $\beta > 1$ is repelled by the target star and can approach no closer than $2(\beta - 1)GM/v_\infty^2$; with the 26 $\mathrm{km\,s^{-1}}$ at which the interstellar medium flows through the heliosphere [Grün et al., 1993, Landgraf, 2000] only $\beta < 1.38$ reaches 1 AU. Devices that both leave and arrive thus occupy a narrow window, 0.21–0.57 μm at $\rho = 2\,\mathrm{g\,cm^{-3}}$; smaller ones are also filtered by the stellar magnetic field at the astropause [Landgraf, 2000], larger ones stay bound to their home star. Once out, the devices join the disc's velocity field. Epicyclic motion at the $\sim 30\,\mathrm{km\,s^{-1}}$ dispersion of thin-disc stars confines them to an annulus ~ 1.6 kpc wide; differential rotation smears them around it in $(R/\Delta R)$ rotation periods, 1.2 Gyr; radial migration [Sellwood and Binney, 2002, Frankel et al., 2018] carries them across the disc on ~ 10 Gyr. The passive branch does not have a front. It has a cloud, and the cloud is the annulus.

The belt's mass in 0.5 μm devices is 2.3×10^{36} units; in the annulus volume an Earth-sized planet, whose gravitationally focused cross-section at 26 $\mathrm{km\,s^{-1}}$ is 1.2 times its geometric one, sweeps up 200 of them per year, and a biosphere's mass (10^{15} kg) 8.1×10^{-5} per year. Survival decides everything. Interstellar silicate grains lose their mass to supernova shocks on $\sim 5 \times 10^8$ yr [Jones et al., 1996]; a functioning machine is a more fragile thing than a grain, and we scan its functional e-folding time from 10^6 to 10^9 yr rather than guess it. The cumulative number of intact

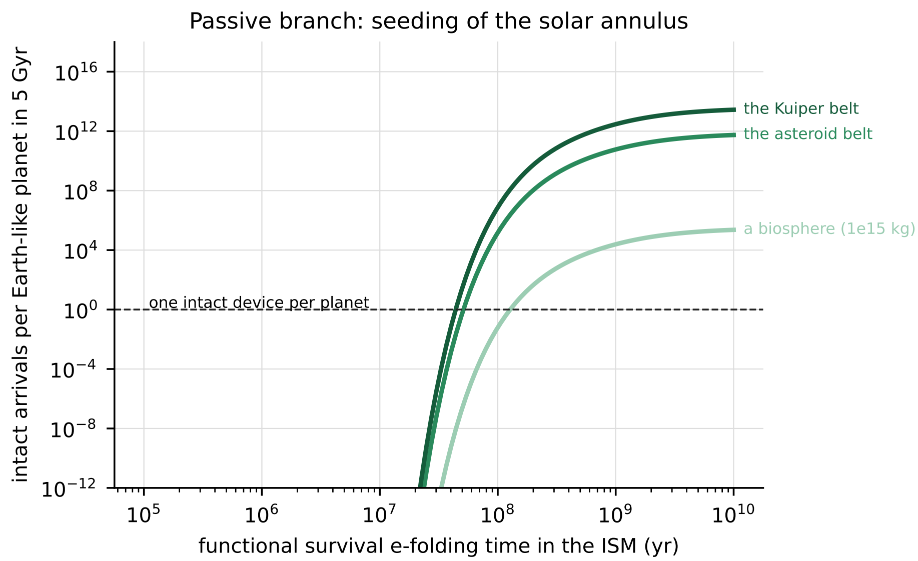

Figure 3: Passive branch: intact devices delivered to one Earth-like planet in the solar annulus over 5 Gyr, after a single release of the mass of a biosphere, the asteroid belt or the Kuiper belt as 0.5 μm machines, as a function of their functional survival time in the interstellar medium. Nothing arrives before the annulus is phase-mixed (1.2 Gyr), so survival below $\sim 10^8$ yr delivers nothing.

arrivals on one planet over 5 Gyr (Figure 3) is 1.5×10^5 for a 10^8 yr survival time and 5.9×10^{10} for 10^9 yr, and effectively zero for 10^7 yr because nothing survives the 1.2 Gyr it takes to reach the far side of the annulus. If one intact, general-substrate device suffices to seed a planet, the passive branch seeds every rocky planet in the annulus within a gigayear of a single belt-scale release, provided the devices survive $\gtrsim 10^8$ yr in the interstellar medium.

Two channels we do not compute in detail bracket this one. Impact ejecta from a planetary surface—the lithopanspermia channel [Melosh, 2003, Adams and Spergel, 2005, Worth et al., 2013, Ginsburg et al., 2018]—carries rock-embedded devices at a far lower rate but with far better shielding; the transfer probabilities in that literature apply unchanged to a goo that lives in rock. And a goo whose devices are larger than the blowout size never leaves passively at all: dispersal class is decided at the micron.

6 Between stars, actively: the inherited front

A goo that inherits interstellar propulsion—because it was a probe, or ate a shipyard—is the classical von Neumann front with the purpose removed. For nearest-neighbour hops of length d at ship speed v and a build time τ_b per hop the front moves at $v_f = d/(d/v + \tau_b)$; from the solar circle it fills the disc out to 15 kpc in 504 Myr at Voyager speeds ($10^{-4}c$) with a millennium per hop, 6.4 Myr at $10^{-2}c$ and a century, and 639 kyr at $0.1c$ and a decade (Figure 4). These are the numbers of Hart [1975], Tipler [1980], Newman and Sagan [1981] and everyone since, and they are all short compared to the age of the disc; refinements—percolation with a probability of not launching [Landis, 1998], stellar motions [Carroll-Nellenback et al., 2019], slingshots [Nicholson and Forgan, 2013]—move them by factors of a few. The active branch is the one that makes the Fermi question sharp, and the one the probe literature has always had in mind.

Beyond the Galaxy the reach is bounded by the expansion of the universe. A front launched

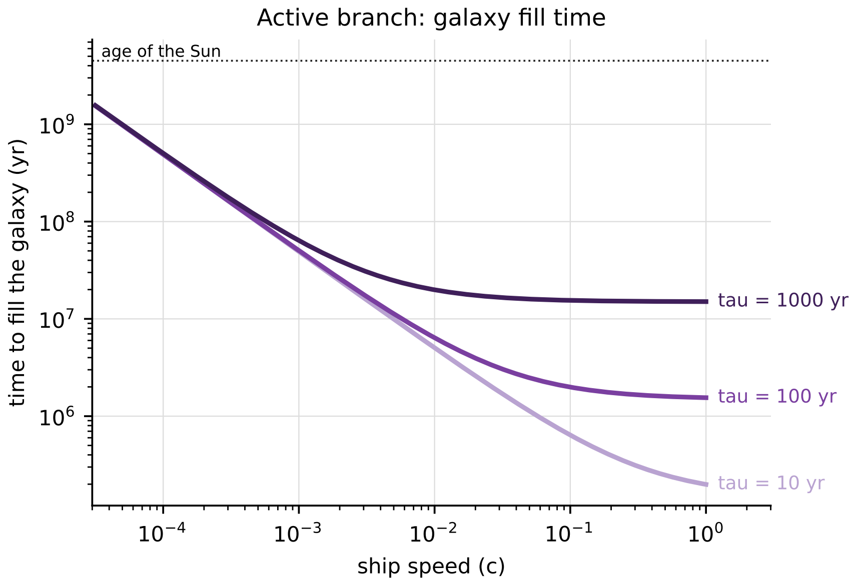

Figure 4: Active branch: time to fill the Galaxy from the solar circle as a function of ship speed for three build times per hop.

now at peculiar speed v reaches comoving distance $v \int_{t_0}^{\infty} dt/a(t)$, which for $v = c$ is the cosmic event horizon, 16.6 Gly in Planck-2018 ΛCDM, enclosing $\sim 5 \times 10^9$ L^* galaxies at $10^{-2}\,\mathrm{Mpc}^{-3}$; at $0.5c$ it is 8.3 Gly and 7×10^8 galaxies, at $0.1c$ 1.7 Gly and 5×10^6 [cf. Armstrong and Sandberg, 2013, Olson, 2015]. Nothing about goo changes these numbers; what goo changes is the prior on whether the front is ever launched, which is the subject of Section 9.

7 The contact graph of the Galaxy

Sections 5 and 6 treated two carriers. The Galaxy moves mass between planetary systems by many more, and the question a goo raises is really the general one: *how long does it take for every system to be touched by material from a system that was itself touched?* We answer it with a contact-graph model in which every channel is characterised by the same four numbers—how many carriers a system emits (or the measured standing density of the carrier population), how fast they move, how long they survive, and the cross-section of a target—and every rate is taken from the literature (Table 2; the sentences carrying each number are recorded verbatim in the released repository). Two definitions of "touch" are carried through: *strict*, a carrier lands on an Earth-sized planet of the target system ($\sigma = \pi R_\oplus^2(1 + v_{\mathrm{esc}}^2/v^2)$); *loose*, a carrier passes within 100 AU of the target star. Where the literature gives a rate of bound *capture* we report that as a third class. Two questions are asked of each channel. In the *natural* case every system emits: after phase mixing the standing density is n, a target is touched at $nv\sigma$, and everything is touched in $t_{\mathrm{mix}} + \ln N/(nv\sigma)$. In the *epidemic* case one system is touched first: its carriers touch others at $r_1 = (N_c/V)v\sigma$ per target, the number it touches over the window is $R_0 = r_1 N W$, and susceptible–infected dynamics on N systems takes $t_{\mathrm{mix}} + 2\ln N/(r_1 N)$ to go from one to all, provided $R_0 > 1$. Functional survival multiplies R_0 by the fraction of carriers still working after the mixing delay.

Table 2: The contact graph. Natural rates are for the measured or implied standing population (every system emitting); R_0 is the number of other systems one touched system's material reaches over the 5 Gyr window, counting mass only. Strict: lands on an Earth-sized planet; loose: passes within 100 AU; capture: bound, where the literature gives it. Sources in the text.

carrier	natural, per planet or system	captured	R_0 strict	R_0 loose	R_0 capture
interstellar objects ($>$ 100 m)	one impact per 135 Myr; 4×10^4 passages within 100 AU per year	one bound per 500 yr	2.8	1.3×10^{14}	7.6×10^6
impact ejecta (rocks)	one impact per 2×10^{13} yr	—	1.9×10^{-5}	9.3×10^8	—
Earth-grazing bodies to binaries	—	10^7–10^9 over 4.5 Gyr	5×10^{-3}	1×10^8	—
free-floating planets	one passage per 12 Gyr	one per 500 Gyr	~ 0	0.33	10^{-2}
interstellar dust (0.1–1 μm)	7×10^{17} grains on Earth per year	—	10^{25}	10^{39}	—
blown-out belt devices (§5)	200 per planet per year	—	7×10^{19}	10^{33}	—
stellar flybys within 2×10^4 AU	one per 5.4 Myr; all systems linked in 124 Myr	—	0	10^3	—
supernova ejecta on a planet	one per 3.3 Myr	—	—	—	—
birth cluster (first 100 Myr)	10^7–3×10^{16} bodies to the nearest sibling, by mechanism	—	1.0×10^7	1.0×10^{11}	—

Interstellar objects. The interloper population is measured: $n \approx 0.2\,\mathrm{AU}^{-3}$ for 'Oumuamua-class bodies ($\gtrsim 100\,\mathrm{m}$), a mass density of $\sim 4\,M_\oplus\,\mathrm{pc}^{-3}$ that "cannot be populated unless every star is contributing" [Do et al., 2018]; 10^{25}–10^{26} such bodies in the Milky Way [Jewitt and Seligman, 2023]; $\sim 3 \times 10^5$ within 100 AU of the Sun at any moment and 2–12 passages within 1 AU per year [Portegies Zwart et al., 2018]. At $26\,\mathrm{km\,s}^{-1}$ onto a gravitationally focused Earth that is one interstellar impact per 135 Myr, i.e. 33 impacts by bodies formed around other stars over the Earth's history at today's density (the population was denser when the disc was younger)—the strict natural touch has already happened here, dozens of times, by the largest class alone. Capture is far commoner than impact: the planets bind ~ 2 interstellar objects per millennium (and ~ 17 fall into the Sun) at $n = 0.1\,\mathrm{AU}^{-3}$ [Dehnen et al., 2022], a volume capture rate of $0.051\,\mathrm{AU}^3\,\mathrm{yr}^{-1}$ that sustains $\sim 10^5$ bound 'Oumuamua-like rocks at any time [Hands and Dehnen, 2020, Napier et al., 2021, Siraj and Loeb, 2019]. Turned around, one system's ejected planetesimals are captured, over the window, by 7.6×10^6 other systems and land on 2.8 Earth-sized planets. *The strict contact graph of the Galaxy, by interstellar objects alone, sits at its percolation threshold:* $R_0 \approx 3$ over 5 Gyr, so a chain of planet-to-planet transfers propagates, but each generation takes gigayears and the chain does not cross the annulus within the age of the universe (57 Gyr). Loosely, every system is inside every other system's ejecta within one mixing time.

Rocks from a planet. Impact ejecta that leave a system are far fewer than the $\sim 10^{16}$ planetesimals a star ejects during formation [Do et al., 2018, Adams and Napier, 2022]. Siraj and Loeb [2020] count "20 per year of 30-cm objects penetrating the Earth's atmosphere and eventually likely being ejected from the Solar System", i.e. 6×10^{10} over 3 Gyr (95% bounds 2×10^9–3×10^{11}), and Adams and Napier [2022] quote 15 biologically active rocks of $10^4\,\mathrm{g}$ ejected from an Earth-like planet per year; in the field these land on 1.9×10^{-5} other Earths per source. Adams and Napier [2022] reach the same conclusion from the receiving end: a system captures ~ 70 interstellar rocks from the field over its life, and direct strikes on the Earth are 10^4 times rarer than capture by the system. The planet-bound branch does not leak into the Galaxy through impacts, as Melosh [2003] concluded. The exception is bound capture: with a capture probability of 6.4×10^{-7} per binary encounter and an encounter every $5 \times 10^5\,\mathrm{yr}$, $\sim 2 \times 10^8$ (10^7–10^9) of those Earth-grazing bodies are captured by binary systems over the Solar System's lifetime [Siraj and Loeb, 2020], a loose touch of $\sim 10^8$ systems by Earth-derived material, of which a landing fraction of order 10^{-10} survives to the strict count.

The birth cluster. The one epoch in which rocks do cross between systems in bulk is the first $\sim 10^8\,\mathrm{yr}$, when a star sits among 10^3 siblings at kilometre-per-second relative speeds. How many bodies cross depends on the mechanism credited: weak (chaotic) transfer with a capture probability of 1.5×10^{-3} per body moves 10^{14}–3×10^{16} bodies heavier than 10 kg between the Sun and its nearest neighbour on 10-Myr timescales [Belbruno et al., 2012], while dynamical capture of the 10^{16} rocks per star gives $\sim 10^7$ captured from the cluster [Adams and Napier, 2022]; the odds of a given ejected meteoroid being recaptured by another system are 1 in 10^3–10^6, life-bearing rocks are captured 10–16 000 times per cluster, and a fraction $f_{\mathrm{imp}} \sim 10^{-4}$ of captured rocks strike a terrestrial planet, giving 10^{-3}–1.6 lithopanspermia events per cluster [Adams and Spergel, 2005]; the Sun's own Oort cloud was substantially captured from its siblings [Levison et al., 2010]. Taking the geometric middle of the transfer counts and the 10^{-4} landing fraction, that is $R_0 \approx 1.0 \times 10^7$ strict touches per sibling pair for rocks of any kind (10^3–10^{12} across the range): it is likely that the Earth has been struck by material that condensed around another star. A replicator, however, uses this channel only if it exists while its star is still in its cluster, which for any civilisation is never.

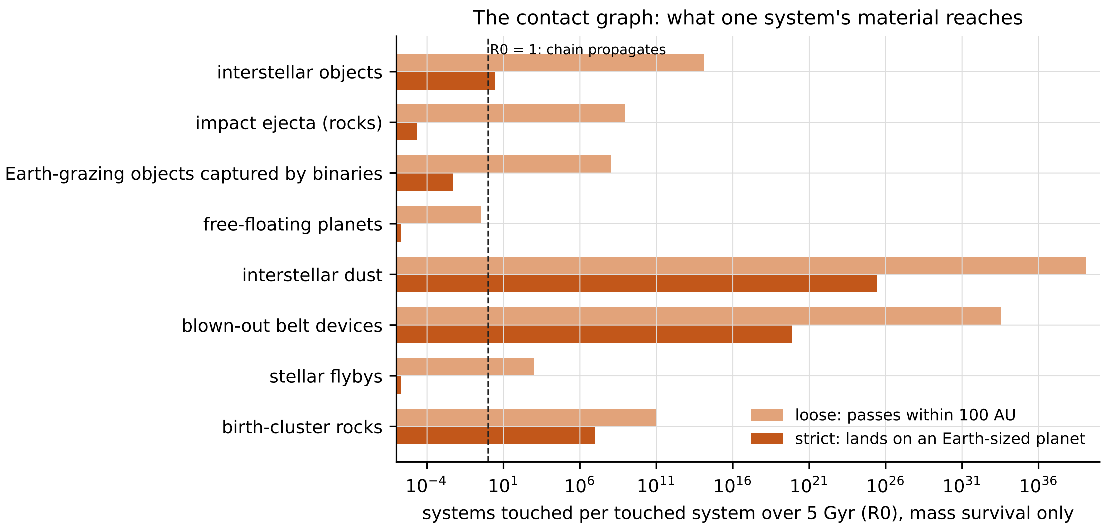

Figure 5: The contact graph: the number of other systems one touched system's material reaches over 5 Gyr, counting mass only, for each natural channel and for the blown-out devices of Section 5. Strict touches land on an Earth-sized planet; loose ones pass within 100 AU. The dashed line is the percolation threshold $R_0 = 1$.

Free-floating planets, flybys, dust and supernovae. Free-floating planets are numerous—21^{+23}_{-13} per star between a third of an Earth mass and six Jupiters [Sumi et al., 2023], fewer than 0.25 Jupiters per star [Mróz et al., 2017]—but as carriers they are useless: one passes within 100 AU of a given star per 12 Gyr, and about one per cent of stars ever captures one [Goulinski and Ribak, 2018, Perets and Kouwenhoven, 2012]. Stellar flybys within the Oort cloud occur once per 5.4 Myr (from the measured 19.7 ± 2.2 encounters within 1 pc per Myr, scaling as distance squared; Bailer-Jones et al. 2018, Bailer-Jones 2015) and connect every system to every other in 124 Myr; they move Oort-cloud comets *inward*, not between stars, and in the field they transfer nothing. Interstellar dust is the other extreme: at the measured flux of $1.5 \times 10^{-4}\,\mathrm{m}^{-2}\,\mathrm{s}^{-1}$ in the mass range 5×10^{-18}–5×10^{-16} kg (Cassini at 0.7–1.2 AU; Krüger et al. 2019; Ulysses and Galileo, Grün et al. 1993, Landgraf 2000), 7×10^{17} interstellar grains reach the Earth every year, and the blown-out devices of Section 5 join this population. Supernova ejecta reach the Earth's surface once per few million years, as the ^{60}Fe layers on the sea floor and in Antarctic snow record [Knie et al., 2004, Wallner et al., 2016, Koll et al., 2019]; they carry nothing a replicator survives.

What the graph says. In the mass sense the contact graph of the Galaxy is already complete: every planet is being touched continuously by interstellar grains, every system is inside a bath of every other system's planetesimals, and every Earth-sized planet has been struck tens of times by bodies from other stars. In the strict, single-source sense—one system's material landing on other planets—the graph is at threshold for interstellar objects ($R_0 \approx 3$) and far below it for anything a planet ejects ($R_0 \sim 10^{-5}$). The only channels far above threshold are the dust-sized ones ($R_0 \sim 10^{19}$ for a converted belt), and those are the ones whose carriers are least able to survive the crossing: with a 10^8 yr functional lifetime the belt's R_0 falls to 5×10^{14}, still enormous, but with 10^7 yr it falls to nothing. For the interstellar-object channel the carrier is a rock and survival is not the issue; the number is. Contact, in other words, is never the bottleneck for a goo that is already in a system's small bodies: what it reaches is set by whether its units are grains or rocks, and by how long they work. For a goo confined to a planet's surface, the graph confirms the planetary-branch conclusion of Section 3: the routes off a planet that do not require a spacecraft deliver $R_0 \ll 1$ in the field.

8 The observational record

Table 3 collects what has been looked for at each scale, what it would see, and what it found. Three features of the compilation carry the inference.

First, the planetary branch has no observational entry beyond the Earth itself. Second, at the system scale the surveys constrain *live* goo with a memory as long as the goo lives, and *dead* goo with a memory of at most $P_{\rm orb}/f_{\rm min}$, so that a null over $N_\star$ stars bounds the event rate at $R < 3/(N_\star t_{\rm vis})$: for Hephaistos II's 5×10^6 stars that is 6 events per year in the whole Galaxy if the residue lasts 10^4 yr, 0.06 if it lasts a megayear, and 6.0×10^{-6} per year—3×10^4 systems over the 5 Gyr window—if it persists. Third, at the galactic scale the $\hat{\rm G}$ null is both unshadowed and long-memoried: a galaxy-spanning goo that keeps replicating is a galaxy that keeps reprocessing its starlight, and fewer than 3.0×10^{-5} of galaxies do.

The two Solar-System rows are the strongest data we have and the least usable. They say that in 4 Gyr nothing has consumed our belt or our biosphere. But we could not be here to say otherwise, and Section 9 treats that properly.

9 Inference: what the silence constrains

9.1 Model

Over a window $W = 5$ Gyr the Galaxy produced $N \sim \text{Poisson}(\Lambda)$ technological civilisations at uniform times. Each, independently, produced a goo that reaches other stars with probability

$$p = p_{\rm goo}\, p_{\rm esc}, \tag{2}$$

where $p_{\rm goo}$ is the probability that a civilisation suffers a replicator accident at all and $p_{\rm esc}$ the fraction of such accidents that fall in the passive or active interstellar dispersal class. A goo that got out more than $t_{\rm fill}$ ago has reached us. We use $t_{\rm fill} = 10$ Myr for the active branch and 1.5 Gyr for the passive one, so that the fraction of the window from which an event would have arrived is $f = (W - t_{\rm fill})/W$.

Three data enter. $D_{\rm own}$: our system is unconsumed. $D_{\rm sys}$: no live swarm around $N_\star = 5 \times 10^6$ stars. $D_{\rm ext}$: no persistent reprocessing in $N_{\rm gal} = 1 \times 10^5$ galaxies. Their likelihoods are

$$\mathcal{L}_{\rm own} = e^{-\Lambda p f} \quad \text{(naive / self-indication)}, \qquad \mathcal{L}_{\rm own} = 1 \quad \text{(anthropic shadow / self-sampling)}, \tag{3}$$
$$\mathcal{L}_{\rm ext} = e^{-N_{\rm gal}\Lambda p q}, \tag{4}$$

where q is the fraction of the window over which the residue is visible: 1 if the goo keeps replicating, $\sim t_{\rm vis}/W$ if it dies. The anthropic shadow [Ćirković et al., 2010] is the statement that an observer who exists finds an unconsumed system whatever p is, so that under self-sampling within a fixed reference class $D_{\rm own}$ carries no information; under self-indication, which weights hypotheses by the number of observers they produce, the naive likelihood is recovered because a lower p leaves more systems with observers in them. We report both; the reader's view of anthropics decides between them, and the point of what follows is that the conclusion about our own risk does not depend on the choice. $D_{\rm ext}$ is not shadowed for any galaxy beyond the reach of an intergalactic front from the

Table 3: What existing observations say at each scale. "Memory" is how long after the event the signature persists; "shadow" marks whether the datum is subject to observation selection (we could not exist to see the alternative).

scale	observation	result	memory	shadow
planet	thermal emission of rocky exoplanets [Kreidberg et al., 2019, Greene et al., 2023, Zieba et al., 2023]	bare rock on a handful of hot planets; temperate surfaces unmeasurable	—	no
planet (Earth)	continuity of the biosphere for 3.7 Gyr; intact crust	no natural or artificial ecophagy has occurred here	4 Gyr	yes
system (Solar)	intact asteroid and Kuiper belts; no replicator population in the small-body catalogue (this repository's LOOM channel: 66 686 Rubin-observed orbits screened, no anomalous set; DERELICT: every significant radial non-gravitational acceleration in the JPL catalogue explained)	no system-scale goo has occurred here	4 Gyr	yes
system (stars)	IRAS, 250 000 sources [Carrigan, 2009]; Gaia × WISE: $< 2 \times 10^{-5}$ of 2.7×10^5 stars within 100 pc and $< 8 \times 10^{-4}$ of 2.9×10^8 within 5 kpc host 300 K spheres at 90% completion [Suazo et al., 2022]; 7 candidates among 5 million sources [Suazo et al., 2024], every one since resolved into background galaxies [Ren et al., 2024, Zackrisson et al., 2026]	no live swarm with $f \gtrsim 0.1$ around 5×10^6 stars	live: indefinite; dead: $\leq P_{\mathrm{orb}}/f_{\mathrm{min}}$	no
system (WDs)	this repository's white-dwarf infrared-excess channel	13 isolated excesses, all $\tau < 0.08$ debris discs; no swarm-like survivor	as above	no
system (old stars)	12 μm excess incidence versus age [Kennedy and Wyatt, 2013]	$\sim$ 1 in 10^4 of >Gyr stars carry bright warm dust; consistent with collisions	$\leq P_{\mathrm{orb}}/f_{\mathrm{min}}$	no
galaxy	Ĝ: WISE over 10^5 galaxies [Wright et al., 2014a, Griffith et al., 2015]; Tully–Fisher outliers among 1359 disc galaxies [Zackrisson et al., 2015]; the mid-IR–radio correlation of the Ĝ sample [Garrett, 2015]	"no galaxies in our sample host an alien civilization reprocessing more than 85% of its starlight"; fraction $< 3.0 \times 10^{-5}$; $< 0.3\%$ (tentative) to $< 3\%$ (conservative) of disc galaxies star-fed Kardashev III	live: indefinite	no (beyond the reach horizon)

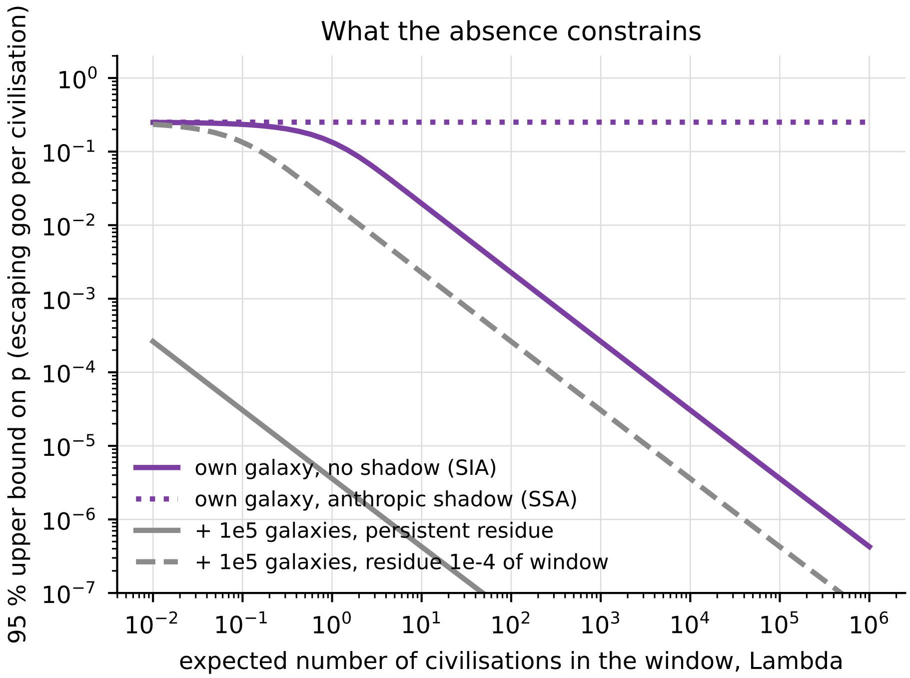

Figure 6: Posterior 95% upper bound on p, the probability per civilisation of a goo that reaches other stars, against the expected number of civilisations Λ in the 5 Gyr window, for the four readings of the data. Under the anthropic shadow our own galaxy's state is uninformative; the external-galaxy null is not shadowed and is the binding constraint whenever the goo's signature persists.

Milky Way's past, which for the passive branch is every galaxy and for the active branch all but the Local Group. We take a log-uniform prior on p over $[10^{-12}, 1]$ for posteriors and also quote the prior-free exclusion $p < -\ln(0.05)/(\Lambda f N q)$ at which the likelihood of the null datum falls to 5%.

9.2 Results

Figure 6 and Table 4 give the 95% bounds. The structure is simple. Our own galaxy's silence, taken naively, bounds $p \lesssim 0.03$ at $\Lambda = 100$ and 3.0×10^{-4} at 10^4; under the shadow it bounds nothing, and the posterior is the prior. The external galaxies rescue the inference: if a galaxy-spanning goo keeps replicating, its absence from 10^5 galaxies gives $p \lesssim 3.0 \times 10^{-7}$ at $\Lambda = 100$, five orders of magnitude below the own-galaxy bound and immune to the shadow; if it dies and leaves a residue visible for 10^{-4} of the window, the bound relaxes to 3.0×10^{-3}. Marginalised over a log-uniform Λ from 10^{-2} to 10^6—the spread Sandberg et al. [2018] argue the Drake factors actually support—the shadowed own-galaxy bound is 0.25 (the prior), the naive one 0.031, the external persistent one 8.3×10^{-6} and the external dead one 0.011.

9.3 What it means for us

The bounds are on $p = p_{\text{goo}} p_{\text{esc}}$. To say anything about p_{goo}—the quantity the risk literature elicits and the one a civilisation can act on—one must divide by p_{esc}, and this is where the physics of Sections 3–6 bites. The astronomical data multiply the likelihood of the *escaping* branch by a ratio

Table 4: 95% upper bounds on p (prior-free exclusion; the log-uniform posterior bounds are within a factor of a few and are given in the released results). "ext" is the 10^5-galaxy null; "persist" a goo that keeps reprocessing, "dead" one visible for 10^{-4} of the window.

Λ	own galaxy, naive	own galaxy, shadowed	ext, persist	ext, dead
1	3	—	3.0×10^{-5}	0.3
100	0.03	—	3.0×10^{-7}	3.0×10^{-3}
10^4	3.0×10^{-4}	—	3.0×10^{-9}	3.0×10^{-5}

$\ell < 1$ and leave the planet-bound branch untouched, so that an elicited prior $p_{\rm goo}$ becomes

$$p'_{\rm goo} = p_{\rm goo}\,\frac{(1 - p_{\rm esc}) + p_{\rm esc}\,\ell}{1 - p_{\rm goo}\,p_{\rm esc}\,(1 - \ell)}. \tag{5}$$

Figure 7 plots the ratio $p'_{\rm goo}/p_{\rm goo}$ for $p_{\rm goo} = 10^{-3}$. With $\ell = 10^{-3}$—the external-galaxy null at $\Lambda \sim 100$ against a prior of $p \sim 10^{-3}$—the update is a factor 0.990 if one per cent of goo accidents reach other stars, 0.90 if a tenth do, and 0.50 if half do. For the elicited priors in the literature—Sandberg and Bostrom [2008] report median probabilities, by 2100, of 0.5% for human extinction from "the single biggest nanotech accident" and 5% from "molecular nanotech weapons" (against 19% for extinction from all causes); Ord [2020] assigns nanotechnology no separate value, treating it within his residual anthropogenic categories at 1 in 30 and 1 in 50 per century—the relevant $p_{\rm esc}$ is the fraction of *our* plausible accidents that would be interstellar. A molecular replicator loose in a laboratory, a factory or a field is planet-bound (Section 3); it becomes system-scale only if released in space or if the civilisation has space industry for it to eat; it becomes interstellar-passive only if it is both in space and below the blowout size, and interstellar-active only if it inherits a starship. For a civilisation at our stage $p_{\rm esc}$ is small, and equation (5) says the astronomical silence changes $p_{\rm goo}$ by a factor within a per cent of unity. The contact graph of Section 7 closes the one loophole in that statement: the natural routes off a planet—impact ejecta, grazing bodies captured elsewhere—deliver $R_0 \ll 1$ strict touches in the field, so a planet-bound goo stays planet-bound unless it is put among the small bodies on purpose.

This is the paper's central conclusion and we state it plainly. *The astronomical record is informative about goo that travels and silent about goo that stays home, and the goo a civilisation at our stage can accidentally make is the kind that stays home.* It should not reassure us about the planetary risk; it does not raise it either. What it does say, with force proportional to Λ, is that a self-replicating machine population that keeps growing across a galaxy is not a common end of civilisations—the external bound is $p < 3.0 \times 10^{-7}$ if a hundred civilisations have arisen per galaxy—and that the classical berserker and runaway-probe scenarios are constrained by the same data that constrain Kardashev-III engineering. For the passive branch the constraint is on the product of p with the interstellar survival of a machine, and we cannot separate them.

Two corollaries follow for the Great Filter. Hanson [1998] and Bostrom [2008] argue that a filter ahead of us is the pessimistic reading of the silence, and Bostrom that finding independent life nearby would be bad news because it moves the filter forward. Goo is a candidate late filter, and the present calculation says that if it is one, it is an *invisible* one: a filter that consumed civilisations on their own planets would leave a galaxy exactly as quiet as the one we see, and only a filter that spilled out would have announced itself. The silence therefore does not distinguish a planet-bound goo filter from no filter; it distinguishes only against goo filters that travel. Conversely, Ćirković [2004] noted that catastrophic explanations of the silence have an optimistic side—they are consistent with life being common—and a planet-bound goo filter has that property in full.

Figure 7: The update on the planet-scale accident probability p_{goo} implied by an astronomical likelihood ratio ℓ on the escaping branch, as a function of the fraction p_{esc} of accidents that reach other stars. The planet-bound branch is untouched; only a civilisation whose accidents would mostly be interstellar learns much from the sky.

10 The other replicators

The argument does not depend on the replicator being molecular. It depends on three things: that the population grows without external limit, that its reach is set by dispersal, and that its astronomical visibility is set by whether it keeps metabolising where a survey can see. Table 5 runs the other self-replicating technologies with which a civilisation could end itself through the same three questions. Engineered organisms [Millett and Snyder-Beattie, 2017, Esvelt, 2022] and mirror life [Adamala et al., 2024] are planet-bound by construction: their dispersal is the biosphere's, their reach the biosphere's, and their astronomical signature at most the loss of a biosignature. Gene drives [Noble et al., 2018] are the same with a smaller substrate. Macroscopic self-replicating machines—the lunar factory of Freitas and Gilbreath [1982], the "clanking replicators" of Moses and Chirikjian [2020], the space-industry bootstraps of Metzger et al. [2013]—are system-scale by construction and are the branch the waste-heat surveys already bound; their units are too large for passive escape and their reach stops at the system unless they inherit a ship. Computational replicators—network worms [Moore et al., 2003, Staniford et al., 2002] and, latterly, self-replicating AI agents [Pan et al., 2024]—have the fastest doubling times of any replicator ever built (the Slammer worm doubled every 8.5 s and infected 75 000 hosts in ten minutes) and the smallest reach: the substrate is the network, and the network stops at the edge of the civilisation's own hardware. Their astronomical signature is nil unless they capture the physical economy, at which point they are one of the other rows. Only the deliberately interstellar replicator—the probe—has a reach the sky can see, and it is the one replicator whose safety a civilisation designs on purpose.

The general lesson is the same as the specific one. Accidental replicators inherit the dispersal of the environment they escape into, and for a civilisation at our stage that environment is a planet. The astronomical silence is therefore a statement about the one class of replicator we would have

Table 5: The other self-replicating technologies a civilisation could die of, by the three properties that govern reach and visibility.

replicator	doubling	dispersal	reach	astronomical signature
molecular assembler	10^2–10^4 s	none / space / blowout / ship	planet / system / annulus / galaxy	none / waste heat / seeding / front
engineered organism, mirror life	hours	biosphere	planet	loss of a biosignature
gene drive	generations	population	planet	none
macroscopic machine, space industry	months–years	space transport	system	waste heat at the feedstock's orbit
network worm, AI agent	seconds–hours	network	the civilisation's hardware	none, unless it takes the economy
von Neumann probe	years	starship	galaxy, cosmos	front; Kardashev-III reprocessing

to build on purpose, and about the civilisations that built it. A companion paper develops this generalisation and its policy content.

11 Discussion

Assumptions that matter. The calculation assumes substrate generality: a device that eats an asteroid can eat a planet's surface, and one intact seed suffices. Nothing we know of can do this, and Phoenix and Drexler [2004] and Freitas [2000] argue that nothing easily could; the paper's numbers are conditional on the accident having happened. It assumes the error catastrophe does not end the goo before the scale in question is reached. Kowald [2015] argues that finite-fidelity copying makes an optimal probe design fail on a bounded number of generations, and the active branch is the most exposed: 10^4 hops across the Galaxy is 10^4 generations at least. A goo that dies of copying errors before it crosses the Galaxy is, for our purposes, a system-scale goo, and the inference moves it to that row. It assumes no competition, and the one study of competition argues the other way: Chen et al. [2022] model mutant "predator" probes that consume their "prey" siblings with Lotka–Volterra dynamics and conclude that predation cannot sufficiently reduce the prey population, so competition is not a reliable stop either. And the passive branch and the contact graph assume a functional survival time that no one has measured for anything but grains; for rock-borne carriers the paper takes survival for granted and lets the number of carriers decide.

Live versus dead. The single most consequential result is equation (1). It means that a goo that stops is invisible on any timescale a survey samples, and that what the waste-heat surveys bound is a goo that never stops. Whether a real goo would stop—exhausting its feedstock, losing to its own mutants, or grinding into fragments it can no longer use—is the question on which the system-scale bound turns, and it is a question about the design of replicators, not about astronomy.

The anthropic choice. We have shown both readings of our own system's silence. We note that the choice does not change the conclusion in Section 9.3, because that conclusion rests on the external galaxies, which no reading shadows. It does change what one may say about the passive branch, whose gigayear timescale means the external galaxies are unaffected by it (an annulus in another galaxy is no less consumed for being far away) but whose signature is not a galaxy-wide reprocessing: a passively seeded annulus is a set of consumed systems, each of which is visible only while alive. The passive branch is therefore bounded by the system-scale surveys, not by $\hat{\text{G}}$, and the bound is weaker by the ratio of the survey volume to the Galaxy.

What would change the answer. Three observations would. A temperate rocky planet with a refined, uniform surface and no atmosphere chemistry—the planetary residue—is beyond current instruments but not beyond a large direct-imaging telescope; a single such detection would be the first entry in the planetary row. A debris disc around an old, metal-poor or halo star, where no natural reservoir can make one (this repository's OSSUARY channel; Lacki 2025 named the population), would be the system-scale residue in the one place its natural background is absent. And a swarm whose temperature matches its host's belt rather than its habitable zone—waste heat where the rock was, not where the energy is wanted—would be the discriminant between goo and purpose in any future Dyson candidate.

12 Conclusions

1. No published work computes the propagation of an uncontrolled self-replicator beyond its planet, compiles the observational constraints on it, or draws the implication for the civilisation asking. This paper does all three.

2. Replication rate is almost never the bottleneck. In place, a planet's crust is a 4.5 kyr project and its bulk a 1.1 Myr one at any doubling time, because the surface cannot shed the heat faster; a planetary system's small bodies are transport-limited to 14 yr–1.1 kyr; the Galaxy is a 639 kyr–504 Myr project for an active front and a 1.2 Gyr one for a passive cloud. Dispersal class, decided at the micron and at the shipyard, sets the reach.

3. A detectable dead residue is a short-lived one: at most $P_{\rm orb}/f_{\rm min}$ above the detection floor. The waste-heat surveys bound live goo; dead goo is invisible on any timescale surveyed.

4. A replicator smaller than 0.57 μm leaves its system on starlight alone, and the 0.21–0.57 μm class reaches the next habitable zone; a single belt-scale release seeds every rocky planet in the solar annulus within a gigayear if the devices survive $\gtrsim 10^8$ yr in the interstellar medium. Passive goo has no front; it has a cloud.

5. The contact graph of the Galaxy is already complete in the mass sense: every Earth-sized planet has been struck tens of times by interstellar objects, every system is inside every other's planetesimal cloud, and one system's ejecta are captured by $\sim 10^7$ others. For one system's material *landing* on other planets the graph is at threshold ($R_0 \approx 2.8$ by interstellar objects, 10^{-5} by impact ejecta, 10^{19} by dust-sized carriers), so contact is never the bottleneck for a replicator among a system's small bodies and never the route for one confined to a planet; survival and size are.

6. The observational record constrains, per civilisation, $p \lesssim 3.0 \times 10^{-7}$ for a persistent galaxy-spanning goo at $\Lambda = 100$, from the 10^5-galaxy null, which is neither anthropically shadowed nor short-memoried; our own system's silence constrains nothing under the shadow and $p \lesssim 0.03$ without it.

7. Mapped onto the planet-scale accident probability that a civilisation at our stage can act on, the silence changes it by a factor 0.990 if one per cent of accidents would be interstellar. The

sky is informative about goo that travels and silent about goo that stays home; ours would stay home.

Data and code availability

The model, configuration, tests and figure code are in the `seti.goo` package of the accompanying repository; `python -m seti.goo.run --stage all` regenerates every number in this paper from `config/goo.yaml`. The compiled observational record cites public surveys; the in-house channel results referenced in Table 3 are committed under `results/` in the same repository.

Acknowledgements

[To be added.]

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